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Astrocyte layers in the mammalian cerebral cortex revealed by a single-cell in situ transcriptomic map

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Although the cerebral cortex is organized into six excitatory neuronal layers, it is unclear whether glial cells show distinct layering. In the present study, we developed a high-content pipeline, the large-area spatial transcriptomic (LaST) map, which can quantify single-cell gene expression in situ. Screening 46 candidate genes for astrocyte diversity across the mouse cortex, we identified superficial, mid and deep astrocyte identities in gradient layer patterns that were distinct from those of neurons. Astrocyte layer features, established in the early postnatal cortex, mostly persisted in adult mouse and human cortex. Single-cell RNA sequencing and spatial reconstruction analysis further confirmed the presence of astrocyte layers in the adult cortex. *Satb2* and *Reeler* mutations that shifted neuronal post-mitotic development were sufficient to alter glial layering, indicating an instructive role for neuronal cues. Finally, astrocyte layer patterns diverged between mouse cortical regions. These findings indicate that excitatory neurons and astrocytes are organized into distinct lineage-associated laminae.

Cortical structure has classically been described in terms of six excitatory neuronal layers, generated sequentially during early development[1]. In addition, the cortex is regionally specialized into areas responsible for motor, sensory and cognitive functions[1]. Although neurons are known to be a diverse population, astrocytes, which comprise about 50% of brain cells, are generally regarded as functionally homogeneous and interchangeable. Might glia also be regionally specialized to confer added cortical organizational complexity? Gray matter astrocytes generally support central nervous system structure, nutrition, regulation of blood flow, the blood–brain barrier, synapse formation and activity[2]. Although several studies show functional diversity of astrocytes[3–7] with implications for regional neural circuit activity or survival[8], precise molecular features of glial spatial organization underlying the cortical architecture remain unclear.

Single-cell spatial transcriptomic approaches can provide unprecedented detail on cellular diversity[9], and have been used to investigate the spatial organization of the mammalian brain[10–12]. Although some approaches produce highly multiplexed gene expression information, imaging and high-content data analysis restrict their application to relatively small tissue areas. Thus, they leave open the important question of how spatial transcriptomics can be applied to screen large tissue areas, including archival tissues and human organs, in a single-cell and quantitative manner to demonstrate gene expression gradients across regions.

In the present study, we developed a method for automated, quantitative, high-content screening of in situ single-cell gene expression in the mammalian brain. We screened 46 candidate genes for astrocyte spatial organization against known patterns for neurons in the cerebral cortex. Surprisingly, although astrocyte expression patterns showed laminar and area organization, such glial layers did not correspond to the six excitatory neuronal layers, indicating higher-order complexity of the cortical architecture.

Results

Automated pipeline to map single-cell cortical gene expression in situ. To map single-cell gene expression across large tissue areas at scale, we automated multiplexed, branched-DNA, single-molecule, fluorescent in situ hybridization (smFISH, RNAScope, Advanced Cell Diagnostics) with a favorable signal-to-noise ratio (SNR)[13], immunohistochemistry (IHC), fast confocal imaging, cellular segmentation and quantification on standard, whole-mouse brain cryosections (Fig. 1a). We used spinning-disk confocal microscopy to image whole-tissue sections at high *XY* and *Z* resolution, then modified a high-content screening workflow software (Harmony, Perkin Elmer) to quantify RNA transcripts per cell. We manually segmented cells

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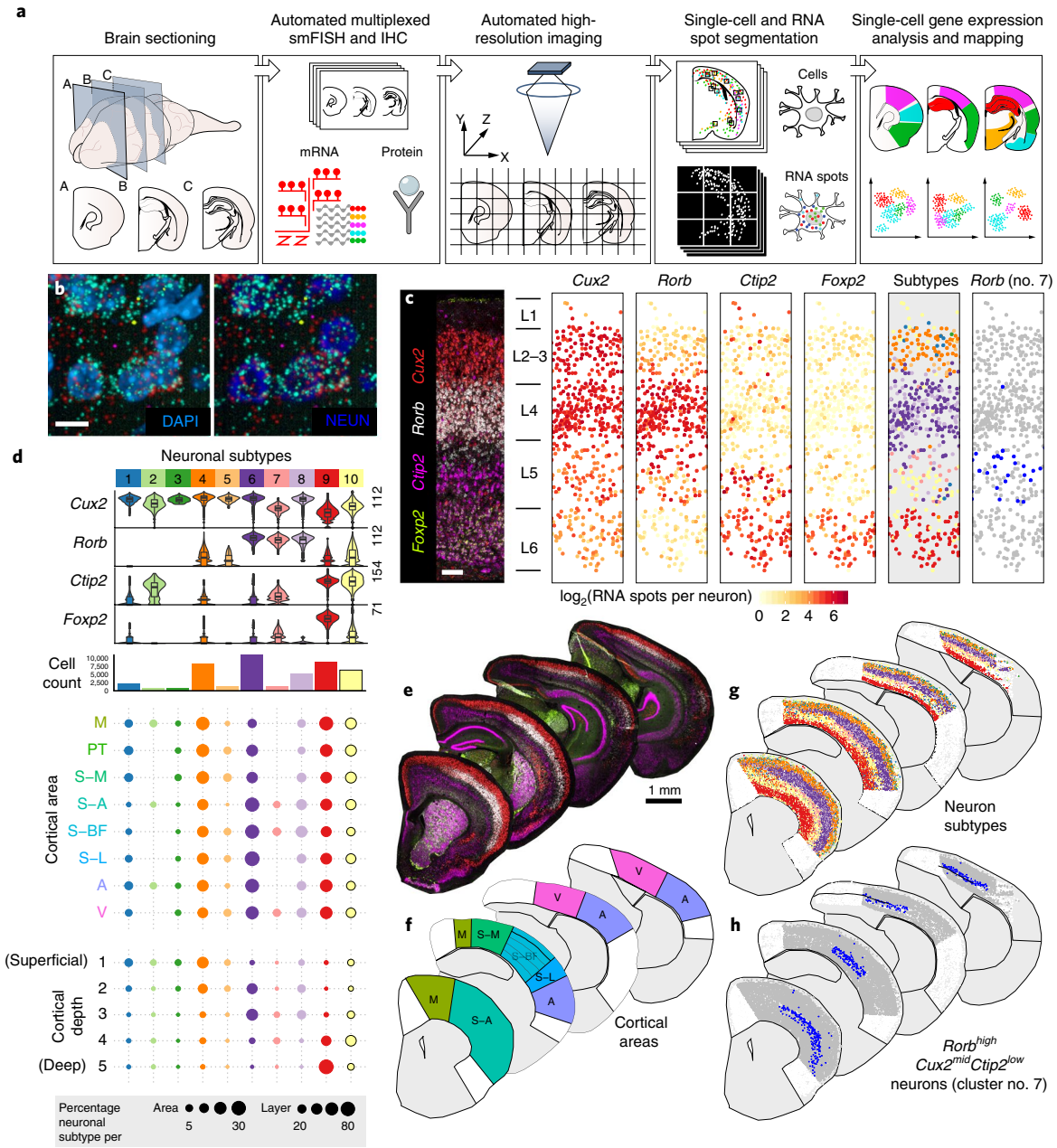


Fig. 1 | LaST map pipeline for mapping cortical neuronal subtypes in situ. a, Design of automated spatial transcriptomic pipeline. **b**, High-resolution imaging of large tissue areas. Shown are ×40 *z*-projection images of *Rorb*$^{+}$ L4 neurons in the P14 mouse barrel cortex. **c**, Automated mapping of layer neuron marker expression and layer neuron subtypes in the mouse barrel cortex. Automatically identified single neurons are plotted as solid circles and colored according to expression (middle panels) or subtype classification (right panels). **d**, Identification of neuronal subtypes based on unbiased classification of single-cell-level smFISH data; *t*-SNE and hierarchical clustering of 46,888 cortical neurons yielded 10 subpopulations. Top: violin plots show single-cell expression profiles of clusters; highest RNA spot counts per cell are shown on the left. Middle: histograms showing total number of cells per cluster. Bottom: spatial distribution of clusters across cortical areas and five normalized cortical depth bins, shown as the percentage of total neurons in given area/depth bin (bottom). **e**–**h**, Single-cell mapping of cortical neuron subtypes. **e**, Low-magnification images of P14 hemisections from four different anatomical levels. **f**, Broad cortical areas included in the analysis. **g**, Maps of 10 major neuronal populations. **h**, Spatial distribution of area-restricted L5 *Rorb*^{high}*Cux2*^{mid}*Bcl11b*^{low} neurons ($n = 1$ mouse, 10 tissue sections independently imaged). Scale bars, 10 µm (**b**), 100 µm (**c**) and 1 mm (**e**). A, auditory; M, motor; PT, parietal; S–A, anterior somatosensory; S–BF, somatosensory barrel; S–L, somatosensory lateral; S–M, medial somatosensory; V, visual.

into brain regions based on reference atlases for gene expression analysis (see Methods) to generate quantitative two-dimensional (2D) tissue maps showing spatial single-cell regulation of gene expression.

LaST map confirms neuron excitatory layering, revealing novel neuronal architecture. We validated our LaST pipeline by mapping expression of known cortical layer neuron markers for layers 2–4 (L2–4, *Cux2*), L4 (*Rorb*), L5–6 (*Bcl11b*) and L6 (*Foxp2*)[1] (Fig. 1b,c) in the mouse brain, followed by automated six-color imaging of 10 anteroposterior sections across the entire P14 cortex (Fig. 1e,f), totaling 2,216 fields of view and ~300,000 images collected over 25 h of continuous acquisition. We used machine-learning-based texture analysis combined with NeuN (neuronal nuclei and cytoplasm) IHC and DAPI staining to identify single neuron soma

on maximum *z*-projection images, while also ruling out doublets/triplets (see Supplementary Fig. 1 and Methods). This approach validated that our technique accurately identified known layer-specific single-cell, and layer-specific neuronal gene, expression patterns (Fig. 1c) with robust reproducibility (see Supplementary Fig. 2).

After generating a quantitative map of known cortical neuron gene expression, we tested whether we could identify and/or discover neuronal subtypes solely from smFISH data in an unbiased manner. We analyzed transcript counts from 46,888 neurons across select broad cortical areas using a modified dimensionality reduction and hierarchical clustering approach (see Methods and Supplementary Table 1). Our analysis identified 10 distinct major clusters based on the expression of the 4 markers mentioned earlier (Fig. 1d) and could robustly detect low expression differences across neurons (see Supplementary Figs. 3, 4 and 5). These clusters primarily represented layer-enriched neuron populations present across multiple cortical areas (Fig. 1c,d,g, for example *Bcl11b*^{high}*Foxp2*^{high} cluster no. 9 in L6). In addition, we captured heterogeneity within cortical layers, exemplified by clusters of *Cux2*^{high}*Rorb*$^{neg/low}$ (nos. 1, 3–5) and *Cux2*^{high}*Rorb*^{high} (no. 8) within L2–3. We could also accurately distinguish area-specific differences such as *Rorb*^{high}*Cux2*^{high} neurons (nos. 6 and 8) enriched in L4 of sensory areas and *Cux2*^{mid}*Bcl11b*^{mid} neurons (no. 2, probably an interneuron subtype, see Supplementary Fig. 6)[14] in L1–3 and L5 of the somatosensory, motor and auditory cortex (full area maps shown in Supplementary Fig. 7). Finally, we found a novel population of *Rorb*^{high}*Cux*^{mid}*Bcl11b*^{low} neurons (no. 7) restricted to L5 of the anterior somatosensory, barrel and visual cortex at P14 and P56 (Fig. 1c,h and see Supplementary Fig. 6). To confirm that *Rorb* and *Bcl11b* expression segregates among neuronal subtypes in L5, we examined available cortical single-cell transcriptome data[15]. Consistently, our survey showed that *Rorb* and *Bcl11b* expression segregates among molecular subtypes of L5 neurons in the adult visual cortex (see Supplementary Fig. 6). Taken together, these results demonstrate that the LaST map can be used to validate regional qualitative and quantitative predictions about gene expression in mammalian brain and can serve as a tool to sensitively identify novel/diversified cell populations in situ.

Laminar gene expression patterns are characteristic of cortical gray matter astrocytes. It is generally accepted that astrocytes in the superficial subpia (L1) and deep L6 layers show white matter (WM) astrocyte characteristics[16]. We next tested whether gray matter astrocytes in L2–5 are molecularly homogeneous or, alternatively, segment into multiple distinct layers. First, to generate candidate gene lists, we manually dissected the upper (L2–4) and deep (L5–6) layers of the somatosensory cortex from P14 astrocyte reporter, *Aldh1L1-GFP* transgenic mice[17] and performed RNA-sequencing (RNA-seq) profiling of FACS-purified astrocytes versus the whole cortex (Fig. 2a). The expression pattern of whole cortical tissue showed that the microdissection accurately captured the upper and deep layers. Astrocyte markers were highly enriched in FACS-purified green fluorescent protein-positive (GFP$^+$) cells over GFP$^-$ cells or whole cortex as expected; moreover, most neuronal layer markers were not distinctly expressed among cortical astrocytes (see Supplementary Fig. 8).

This analysis identified 159 candidate genes with a differing expression between the upper- and deep-layer astrocytes (Fig. 2b). Although most genes showed moderate-layer enrichment, many also showed layer- and astrocyte-enriched expression in published cortical transcriptome datasets (see Supplementary Fig. 9). We next validated layer astrocyte gene expression in situ. To distinguish astrocyte-specific gene expression by smFISH, we examined markers for astrocytes (*Aldh1l1*, *AldoC*, *Glast*), neurons (*Syt1*, NeuN), oligodendrocytes (*Pdgfra*, *Plp*), microglia (*Tmem119*) and endothelia (*Tie1*)[17]. As shown in Fig. 2c–g, although *Glast* messenger RNA showed low expression in *Pdgfra*$^+$ oligodendrocyte precursors, high *Glast* levels exclusively marked astrocytes. *Glast* mRNA (but not *AldoC* or *Aldh1L1*) filled astrocyte soma and the main processes, permitting clear delineation of astrocyte cell borders by image texture analysis.

To avoid double counting, we selected non-overlapping single astrocytes by filtering cells based on size, DAPI (single nuclei) and *Glast* intensity (see Methods, and see Supplementary Figs. 10 and 11). In addition, we filtered astrocytes and their processes that overlap with neuronal soma and non-astrocyte nuclei in *z*-projection images (Fig. 2c, and see Methods). These measures ensured that our smFISH approach counted single-cell, astrocyte-specific, gene expression in situ in the context of a large tissue area screen.

To validate candidate layer-associated astrocyte genes in situ, we first analyzed the P14 somatosensory barrel cortex. We selected the top 46 genes from RNA-seq that showed the highest differential enrichment between cortical layers and/or astrocyte specificity (Fig. 2h, and see Supplementary Table 2). Of these, we confirmed layer-enriched expression patterns or rejected genes that showed subtle variations based on quantitative findings in situ. For example, 48% (22/46) showed an expression pattern biased toward—but not restricted to—the upper layers (for example, *Tnfrsf19*, *Mfge8*), and 15% (7/46) showed enrichment in all gray matter astrocytes across L2–6 (for example, *Igfbp2*, *Kirrel2*). Another third (13%, 6/46) showed expression in astrocytes + neurons and were excluded (for example, *Ddit4l*, *B3galt2*; see Supplementary Fig. 12). In contrast, eight genes clearly distinguished upper- versus deep-layer astrocytes in the somatosensory cortex (Fig. 2h). As shown (Fig. 3a–c, and see Supplementary Fig. 13), expression of the bone morphogenetic protein antagonist chordin-like 1 (*Chrdl1*)[18] was expressed in L2–4 astrocytes, whereas interleukin-33 (*Il33*)[19] was enriched in L5–6. In distinction, sciellin (*Scel*), a component of epithelial cornified envelopes[20], was expressed in mid-cortical (~L4) layers and transcriptional repressors *Id1* and *Id3* occupied deep (L5–6) and marginal (L1) layers[21]. These findings show that subsets of cortical gray matter astrocytes show laminar spatial gene expression heterogeneity.

Astrocyte layers diverge from neuronal laminae. To determine whether glial layers matched excitatory neurons, we directly compared the spatial expression patterns of the eight top-layer astrocyte markers against layer neuron markers in P14 somatosensory barrel cortex in situ. We found that astrocyte layer genes generally fell into three spatial bins comprising superficial, mid and deep laminae. As shown (Fig. 3a–d, and see Supplementary Fig. 13), *Chrld1*, *Scel*, *Eogt*, *Spry1*, *Paqr6* and *Il33* were expressed in patterns that spanned multiple neuronal layers. Astrocyte *Chrdl1* expression peaked in L2–4, with low levels in L1 and L5. *Scel* expression was highest in L4–5, with an upper boundary in mid L2–3. *Il33* expression peaked in L5, with low levels in L4 and L6. Such patterns were not artifacts of astrocyte size or *Glast* expression level differences across layers (see Supplementary Fig. 14) and were highly reproducible, with expression levels similar to those of layer neuron markers (see Supplementary Fig. 15). The layer astrocyte patterns were also validated with IHC (see Supplementary Fig. 16). Moreover, analysis of the P56 mouse brain indicated persistence of these patterns, except for the mid-layer *Scel* expression into adulthood (Fig. 3d). Indeed, we found that the upper-layer astrocyte *Chrdl1* expression was conserved in the adult human brain (Fig. 3e, and see Supplementary Fig. 17).

Although we found certain similar features between astrocytes in deep cortical layers and WM, most cortical astrocyte layer genes were specific to gray matter. L6 astrocytes expressed high levels of *Gfap*, *Id1* and *Id3*, similar to astrocytes in subcortical WM and L1 subpia (see Supplementary Fig. 13)[7]. Conversely, *Il33* was enriched in L5–6 gray matter astrocytes but absent from WM, showing that deep-layer astrocyte identity is distinct from WM. Together, these

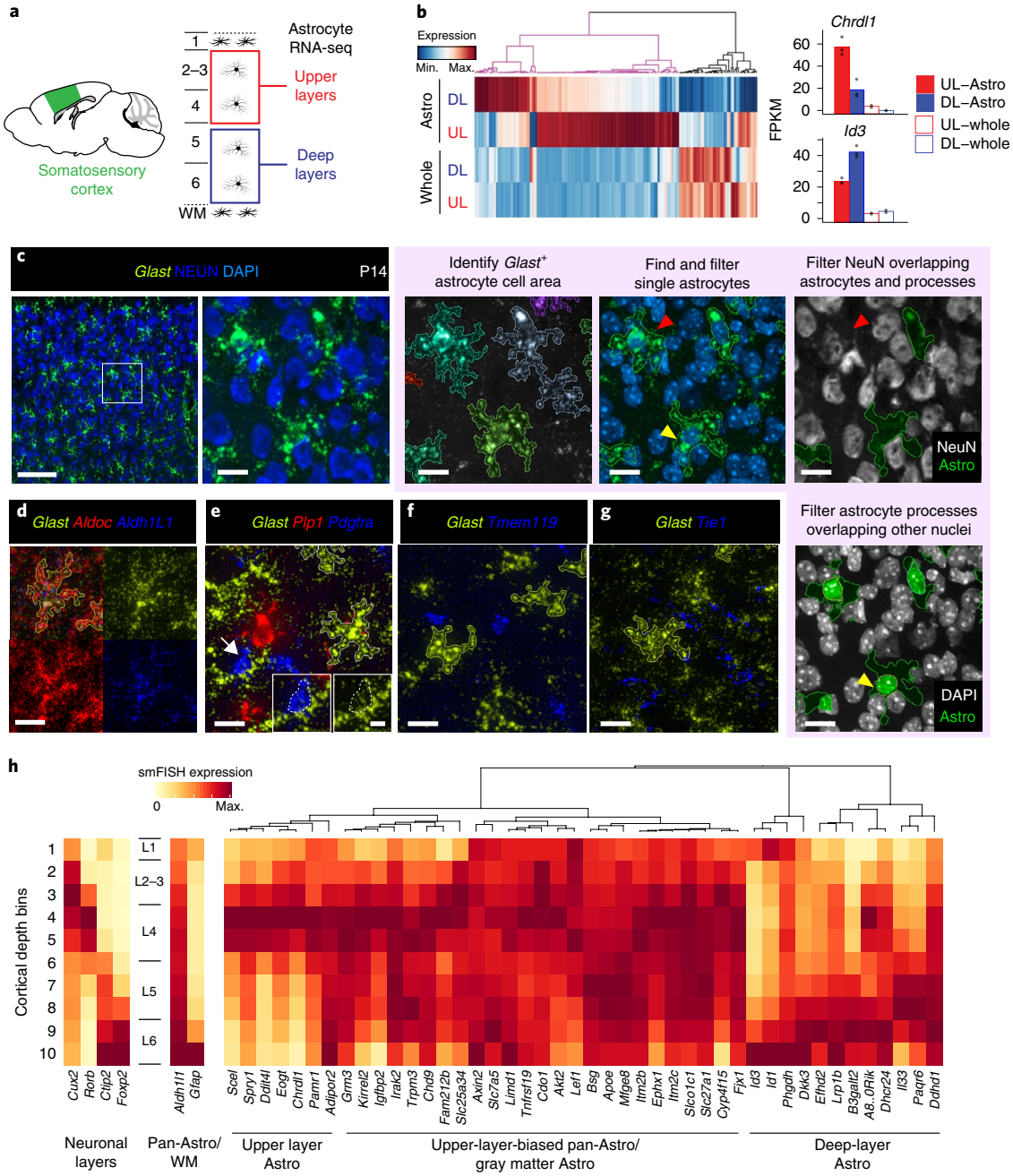


Fig. 2 | Novel layer expression differences among cortical gray matter astrocytes revealed through RNA-seq and LaST map. a, Diagram showing somatosensory cortical areas and layers used for laminar gray matter astrocyte RNA-seq expression profiling. **b**, Novel molecular heterogeneity of layer astrocytes (Astro) identified by RNA-seq: 159 differentially expressed genes between upper- (UL) and deep-layer (DL) gray matter astrocytes (FDR < 0.05 and expression threshold >5 FPKM) were detected. Bar plots show mean expression ($n = 3$ mice for astrocytes; $n = 2$ mice for whole cortex). **c**, Identification of astrocyte gene expression with *Glast* (*Slc1a3*) smFISH. The astrocyte cell area is segmented from the *Glast* signal and nuclei are segmented from DAPI. Single astrocytes are selected using morphologic and intensity filters; non-astrocyte nuclei and overlapping neurons are excluded. **d**–**g**, *Glast* is a specific marker of astrocytes. Solid outlines indicate the cell areas of identified single astrocytes. *Glast* expression was assayed with markers of astrocytes (**d**), oligodendrocytes (**e**), microglia (**f**), and endothelia (**g**). **h**, Screening of candidate layer astrocyte genes with LaST map identifies laminar expression patterns in situ. Tile heatmaps show average single-cell gene expression binned across cortical depth in the P14 somatosensory cortex ($n = 2$ pooled biological replicates across multiple tissue sections). Upper- and deep-layer genes with astrocyte-specific expression are marked in bold ($n = 2$ mice independently assayed, three tissue sections imaged per replicate). Scale bars, 100 µm (**c**, left panel), 25 µm (**c**, other panels), 25 µm (**d**–**g**) and 10 µm (**e**, inset).

findings indicate that astrocytes are organized into multiple layers across the cortical gray matter and that several glial laminae are dissimilar to classic neuronal layers (Fig. 6e). Moreover, such patterns are evident from P14, persist until adulthood and are conserved in the human cortex.

Astrocyte layers are observed in unbiased single-cell transcriptomic data. To independently verify astrocyte layers, we examined the molecular heterogeneity of cortical astrocytes in a single-cell RNA-seq (scRNA-seq) study (Fig. 4a). The single-cell astrocyte transcriptomic database was generated using a modified Smart-Seq2

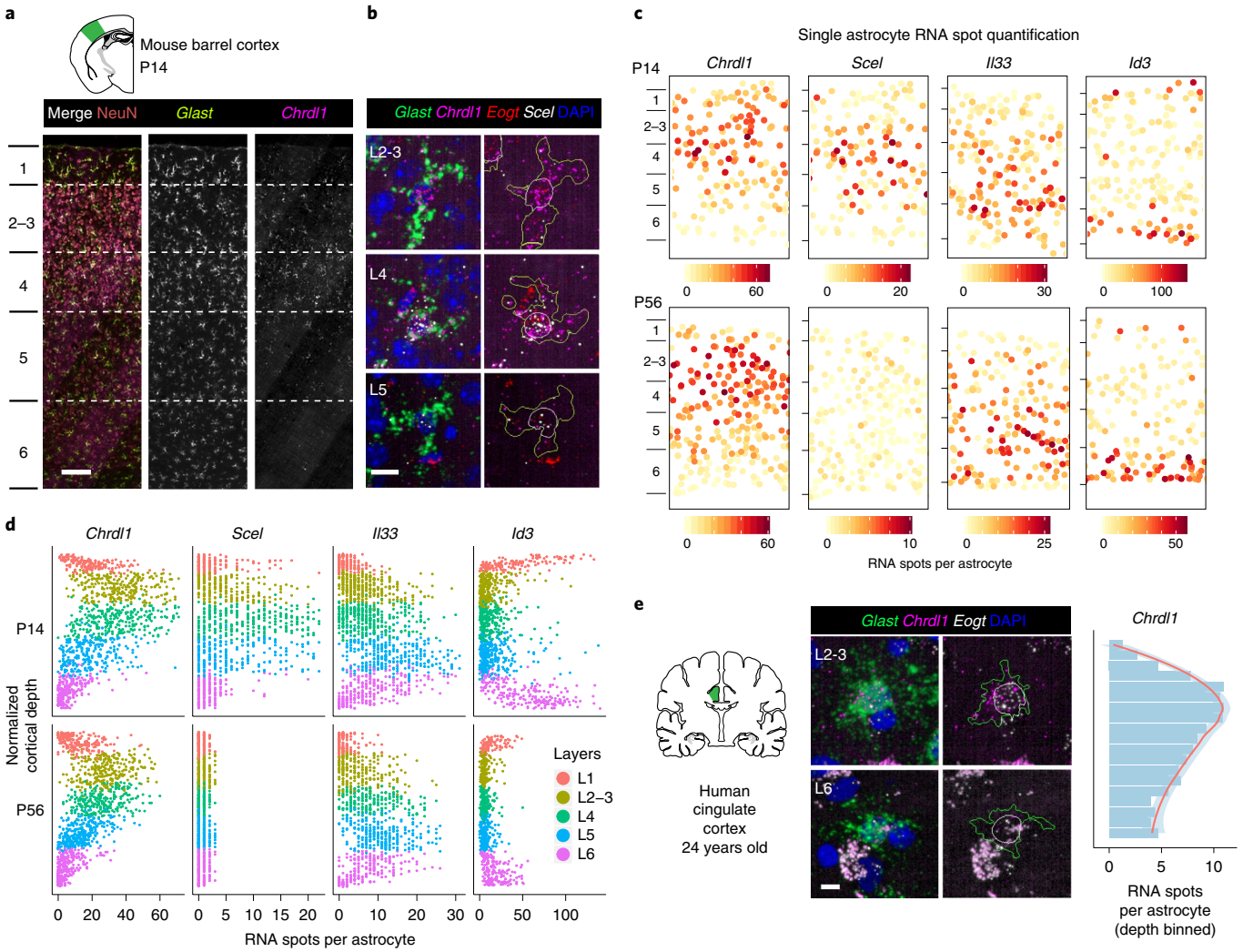


Fig. 3 | Astrocytes show broad expression gradients across cortical depth and diverge from neuronal layers. a,**b**, Upper-layer astrocyte enrichment of *Chrdl1*, *Scel* and *Eogt*. Images show the mouse barrel cortex at P14 (**a**) and close-ups of astrocytes across layers (**b**). Astrocyte cell areas are marked with yellow outlines and nuclei are marked with white outlines. **c**, Astrocytes are organized into superficial, mid and deep layers across the cortical gray matter. Single astrocyte expression maps in the barrel cortex at P14 (top panels) and P56 (bottom panels). Astrocytes are plotted as solid circles and colored quantitatively for RNA spot counts per gene per cell. **d**, Single astrocyte quantification of layer astrocyte markers across cortical depth in the barrel cortex at P14 and P56. **e**, Upper-layer astrocyte enrichment of *Chrdl1* expression in the adult human cingulate cortex. Quantification of depth-binned, average single astrocyte expression is shown on the right ($n = 1$ mouse per time point; three tissue sections imaged independently per gene panel (**a**–**d**) and three human brains independently assayed; one to two sections imaged per case (**e**)). Scale bars, 100 µm (**a**), 10 µm (**b**,**e**).

protocol[22]. Unsupervised clustering with the Louvain method identified two major astrocyte subtypes in the P56 cortex (Fig. 4c). These astrocyte clusters, designated AST2 and AST3, were distinguished by the expression of *Unc13c* and *Agt*, respectively (Fig. 4c), and were found to be largely intermixed throughout cortical layers[22]. To reconcile our results with these findings, we examined the expression pattern of our cardinal layer astrocyte markers in the scRNA-seq data. As shown (Fig. 4d), our layer astrocyte markers were expressed in subpopulations of cells in the single astrocyte transcriptome data, but not in a cluster-associated way. These observations suggest that, although layer astrocyte gene expression patterns can be detected in an unbiased transcriptomic study of cortical astrocytes, there might be multiple axes of astrocyte heterogeneity (that is, laminar and non-laminar) in the cortex.

To determine whether astrocyte layering represents an axis of heterogeneity in the single-cell transcriptome data, we then mapped the scRNA-seq data to cortical layers using the smFISH data as a spatial reference (Fig. 4b). To infer the cortical layer position of cells profiled in scRNA-seq, we employed a Bayesian model[23] using 16-layer astrocyte markers profiled with LaST map smFISH as a reference. The spatial reconstruction allowed us to assign cells to different layers with high accuracy (Fig. 4f, and see Supplementary Fig. 18) and distinguish astrocyte layers in the single-cell transcriptome data (Fig. 4e). The two cortical astrocyte subtypes identified by unsupervised clustering of scRNA-seq were both located across multiple layers (Fig. 4e, and see Supplementary Fig. 19), consistent with the observation that they are spatially intermixed[22]. We also found that scRNA-seq clusters derived using the alternative SC3 method showed a similar non-layer-specific distribution (see Supplementary Fig. 19). Taken together, these results suggest that cortical astrocytes show at least two axes of molecular diversity: (1) layer-independent heterogeneity detected by unsupervised clustering of scRNA-seq data and (2) layer-dependent heterogeneity demonstrated by our smFISH-based approach and the spatial reconstruction of scRNA-seq data.

Spatial reconstruction of single astrocyte transcriptomes identifies new layer astrocyte genes. To expand the repertoire of astrocyte layer markers, we used spatial reconstruction of single-cell transcriptomes[23]. This analysis yielded 125 genes with layer-associated expression features. Layer-associated astrocyte candidate genes comprised, on average, 19% of transcripts per cell and 8% of expressed genes (se Supplementary Fig. 18, Kruskal–Wallis false discovery rate (FDR)-corrected $P < 0.05$); of these, two-thirds were predicted to exhibit upper- and deep-layer astrocyte expression (similar to *Chrdl1* and *Il33*), whereas the remaining third of genes exhibited WM-like layer profiles (similar to *Id3*; Fig. 4g, and see Supplementary Fig. 18). We validated three of these predictions with smFISH. As shown (Fig. 4h–j), the expression of the transcriptional regulator nuclear protein 1 (*Nupr1*) peaked in L2–4 astrocytes, whereas the enzyme epoxide hydrolase 2 (*Ephx2*) and the proteolipid neuronatin (*Nnat*) showed higher expression in deep-layer L5–6 astrocytes. Thus, interrogation of scRNA-seq data provides additional insight into the degrees of astrocyte layer heterogeneity in the cerebral cortex.

Neuronal specification directly or indirectly determines astrocyte laminar gene expression. During cortical development, neuronal layers are established along a radial glial scaffold before astrogenesis (postnatal days 1–7 (P0–7))[24]. To investigate whether neuronal factors regulate astrocyte layer organization, we first examined mice that lack the chromatin remodeling factor *Satb2*, required to specify superficial post-mitotic callosal neuron identity[25]; *Satb2* conditional knockout (*Satb2* cKO) mice (*Emx1-cre* × *Satb2*$^{floxed/floxed}$) are deficient in L2–4 neuron gene expression, lack callosal axon projections and ectopically express markers of L5–6 neurons in superficial cortical layers[26,27].

We verified that, at P14, L4 neurons (*Rorb*, *Pamr1*) are absent in *Satb2* cKO mice whereas L2–3 neurons show some ectopic deep-layer gene expression (*Bcl11b*) (Fig. 5a,c, and see Supplementary Fig. 20). As a consequence, we found complete loss of *Scel* expression in astrocytes, whereas upper-layer astrocyte genes *Chrdl1* and *Eogt* were significantly reduced (Fig. 5b,d). Deep-layer astrocyte genes (*Il33*, *Id1*, *Id3*) were unaffected. These findings indicate that a post-mitotic, L4-specific neuron identity is necessary for the generation of superficial-layer astrocyte identity (Fig. 5i). Next, to assess deep-layer astrocyte identity in relation to neurons, we used *Reeler* (*Reln*) mice, deficient for secreted reelin, which show an inversion of radial glial polarity and neuronal layers[28]. In *Reln* mice excitatory neurons are specified correctly but fail to migrate to their proper layers, yielding an 'inside-out' cortex before birth[29]. As expected[30], at P14 the expression pattern of superficial (*Cux2*) and deep laminar markers (*Bcl11b*, *Foxp2*) were inverted, whereas L4 neurons (*Rorb*) formed aberrant clusters across cortical depth (Fig. 5e–g). In addition, we found that superficial and deep astrocyte layers were inverted (Fig. 5f–h). Consistent with the L4 neuron phenotype, *Scel*$^{+}$ astrocytes were aberrantly located across cortical depth. Together, these findings suggest that cortical neuronal cues are involved in the regulation of astrocyte layer development (Fig. 5i).

Astrocyte layer characteristics vary between cortical areas. Are astrocyte layers distinct or similar between functional cortical areas? To investigate astrocyte arealization, we quantified patterns of top-layer astrocyte markers across the P14 cortex. As shown (Fig. 6), astrocyte layer gene expression varied notably across the dorsoventral and rostrocaudal cortex. *Scel* expression showed enrichment in L4–5 astrocytes in the somatosensory areas, but was absent from the medial motor and caudal visual cortex (Fig. 6a,b), a pattern that correlates closely with L4 neurons (Fig. 6d). In a different pattern, *Chrdl1* showed dorsoventral enrichment in L1–4 astrocytes throughout the cortex (Fig. 6c). In contrast, deep-layer astrocyte marker expression was enriched in medial and lateral areas across the dorsoventral axis (*Il33, Id3*, Fig. 6d, and see Supplementary Figs. 21 and 22). Thus, astrocytes show both laminar and area heterogeneity across the cortex. Collectively, our observations on the regional heterogeneity of cortical astrocytes and their induction by neuronal signals show that glial laminae are a fundamental biomarker characteristic of cortical layer and area architecture (Fig. 6e,f).

Discussion

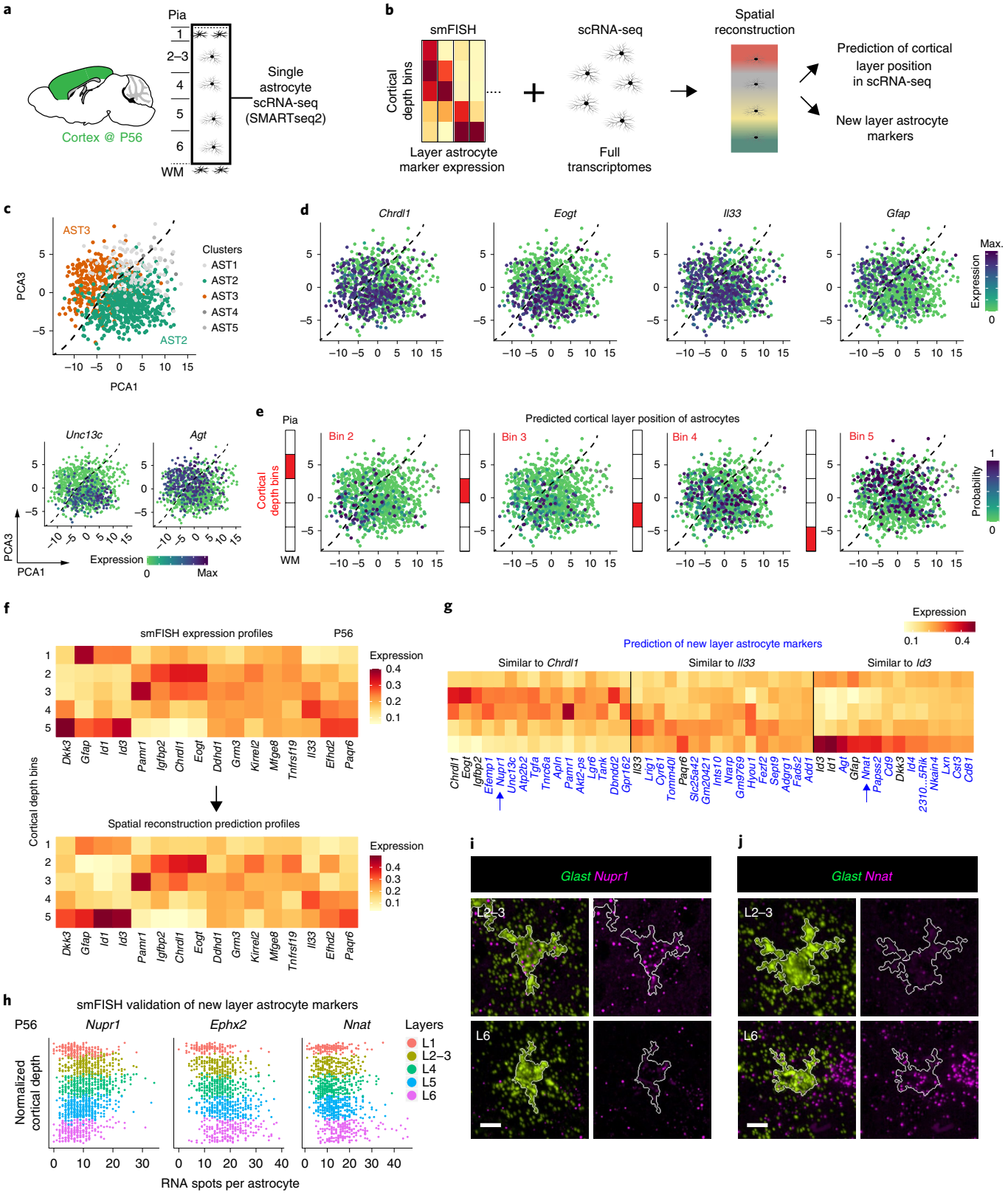
Our findings indicate LaST map as a robust technique to validate single-cell gene expression in situ as well as discovery of combinatorial patterns indicative of cell diversity. For example, our findings indicate a novel *Rorb*, *Bcl11b* L5 neuronal subtype in visual cortex, as well as molecular diversity of cortical gray matter astrocytes. Although a previous study indicated morphologic and gene expression differences in gray matter versus WM/L6 cortical astrocytes[7], a surprising biological finding of the present study is that cortical gray matter astrocytes are organized into laminae that are distinct from classic layers of excitatory neurons. Moreover, the present study identified distinct astrocyte subtypes within gray matter L2–5. Single-cell quantification of astrocyte layer genes uniquely showed peaks and troughs of expression across the superficial-to-deep cortex, as well as differences across functionally distinct cortical areas. As previous studies show that region-restricted astrocyte mRNA and protein expression are predictors of functions tailored to support local neural circuits[8,31,32], astrocyte laminar genes indicate potential additional localized functions. Indeed, a recent study showed that expression of *Chrdl1* in upper-layer astrocytes is essential for local synapse regulation and regulation of developmental plasticity[33].

By introducing an in silico spatial reconstruction approach, we identified and validated further markers of layer-restricted astrocytes in situ. This indicates that identification of cell type- and region-specific markers by LaST map can be used to interrogate other datasets for insights into diversified function, which could play into sophisticated layer-associated neural circuit activity[34]. Interestingly, in a companion analysis, genes from cortical

Fig. 4 | Spatial reconstruction of astrocyte layers from single-cell transcriptome data. a,**b**, Diagrams showing cortical areas and layers used for single astrocyte RNA-seq expression profiling (**a**) and strategy for spatial gene expression reconstruction (**b**). **c**, Astrocyte (AST) clusters in scRNA-seq data visualized using principal component analysis (PCA) plots (PC1, 3%, and PC2, 0.77% of variance). Dashed lines indicate the border between the major clusters AST2 and AST3 (colors, top). Bottom plots show the expression of cluster markers ($\log_2$(Smart-seq2 read counts); $n = 2$ independent experiments). **d**, Astrocyte layer markers express in subpopulations of astrocytes in scRNA-seq data but not in a cluster-specific manner. PCA plots are shown. **e**, Predicted astrocyte layering across scRNA-seq data as a new axis of heterogeneity. PCA plots (*x* axis is PC1, *y* axis is PC3) indicate the probability of cell assignment (color) to the cortical depth bins as shown. **f**, The smFISH reference of 16-layer astrocyte markers used for the reconstruction (top) and the predicted output of the reconstruction (bottom). Expression was scaled to sum to 1 across bins (*y* axis). **g**, New candidate layer astrocyte genes predicted by the spatial reconstruction model. The expression pattern of the top 10 new layer astrocyte genes (*x* axis) that are most similar to *Chrdl1*, *Il33* and *Id3* are shown. Expression was scaled to sum to 1 across bins (*y* axis). **h–j**, Validation of three new candidate layer astrocyte genes using LaST map smFISH. Single astrocyte quantification across cortical depth in the barrel cortex at P56 ($n = 3$ pooled tissue sections from one replicate per time point) (**h**). **i**,**j**, Close-up images of astrocytes. *Nupr1* is enriched in upper layer astrocytes (**i**) while *Nnat* shows enriched expression in deep layer astrocytes (**j**). White outlines mark astrocyte cell areas (**i**,**j**) ($n = 2$ mice independently assayed, three sections imaged). Scale bars, 10 μm (**i**,**j**).

astrocytes self-organized into two classes with a largely excitatory or inhibitory synapse character across layers[22]. Together, these findings suggest that astrocyte laminae include a further axis of heterogeneity in terms of interactions at the synapse level in excitatory or inhibitory circuits.

International collaboratives such as the Human Cell Atlas and BRAIN initiative have generated large repositories of gene expression data, and a logical next step is to map back such expression patterns in situ within developing tissues and/or mature organs. However, spatial transcriptomic analysis in human tissue presents



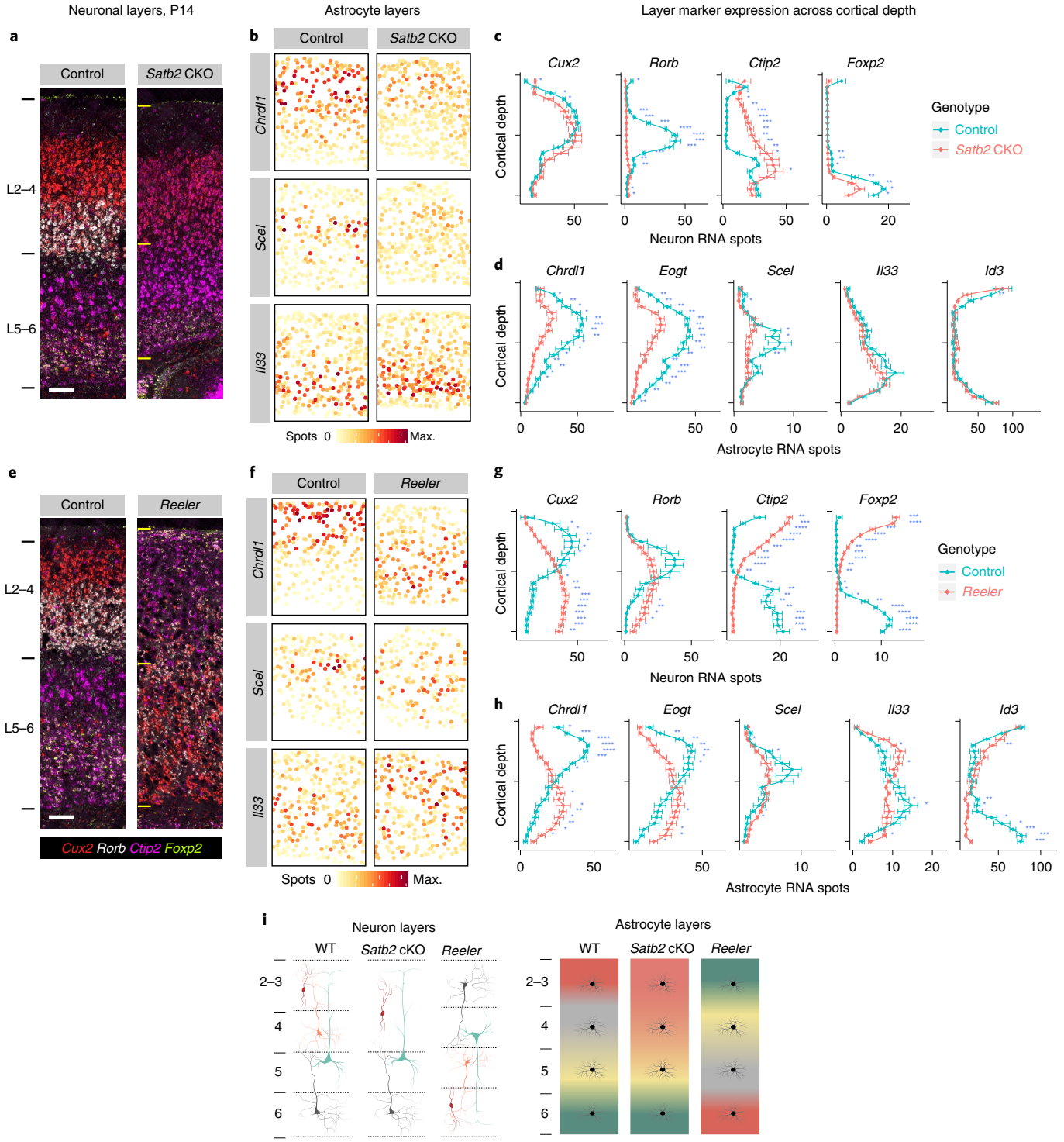


Fig. 5 | Evidence that post-mitotic neuronal cues establish astrocyte layer identities. a–d, *Satb2* cKO mice show defects in upper-layer neuron and astrocyte identity. **a**, Images showing aberrant upper neuronal layers in the *Satb2* cKO barrel cortex at P14. **b**, Single astrocyte maps of layer astrocyte marker gene expression. **c**,**d**, Quantification of cortical depth-binned layer neuronal (**c**) and astrocyte (**d**) marker expression in cKO versus control mice. **e–h**, *Reeler* mice show inversion of neuronal and astrocyte layers. **e**, Images showing neuronal layer inversion in the *Reln*$^{-/-}$ barrel cortex at P14. **f**, Single astrocyte maps of layer astrocyte marker gene expression. **g**,**h**, Quantification of cortical depth-binned layer neuronal (**g**) and astrocyte (**h**) marker expression in cKO versus control mice. **i**, Diagrams depicting layer neuron and astrocyte changes in *Satb2* cKO and *Reln*$^{-/-}$ mice ($n = 3$ mice per genotype independently assayed, 5 tissue sections from each replicate imaged). Scale bars, 100 µm. All data represent mean ± s.d. Two-tailed Student's *t*-tests were used; $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$.

a particular scaling challenge. LaST map provides a pipeline to map quantitative gene expression in single cells across large mouse and human tissue areas. It uses commercially available probes and does not require special tissue preparation or bespoke imaging equipment. LaST map showed reproducibility across multiple tissue samples/sections and should be adaptable for a range of other tissues

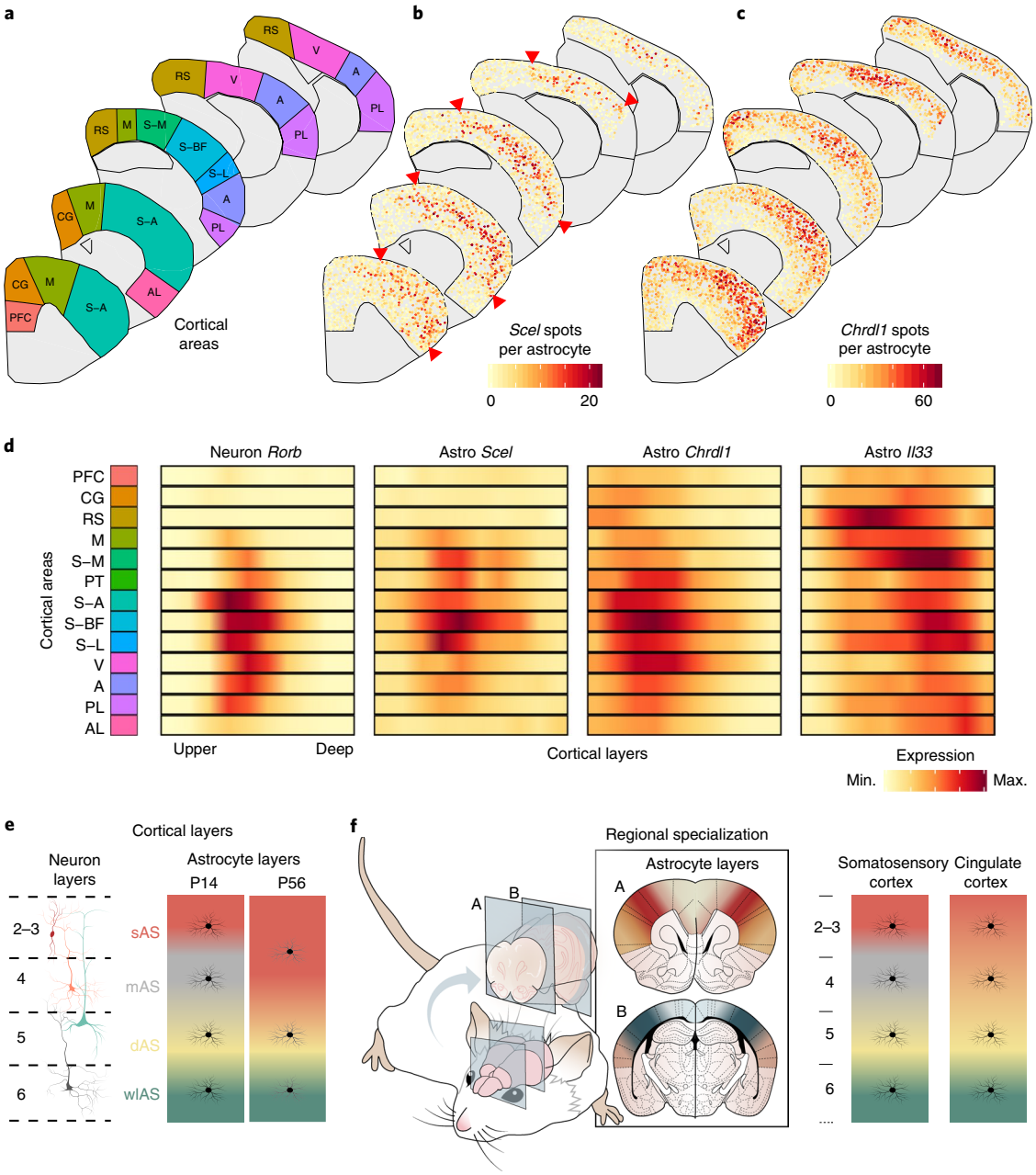


Fig. 6 | Astrocyte arealization across the cortex. a–c, Single-cell mapping of astrocyte gene expression across cortical areas. Maps show the P14 cortical areas used in the analysis (**a**), single astrocyte expression of *Scel* (**b**) and *Chrdl1* (**c**) across the dorsoventral and rostrocaudal extent of the cortex. Astrocytes are plotted as solid circles and colored quantitatively for RNA spot counts per gene per cell. Arrowheads indicate the restriction of *Scel* expression to sensory areas. Abbreviations are as in Fig. 1. **d**, Astrocyte layers are distinct across cortical areas. Smoothed tile plots show the quantification of neuronal *Rorb* expression and astrocyte expression of *Scel*, *Chrdl* and *Il33* across the cortical areas and layers. **e**, Diagrams showing the divergent layer heterogeneity of astrocytes. The present study identifies superficial (sAS), mid (mAS) and deep (dAS) layer astrocyte subtypes through cortical gray matter in the postnatal cortex, and confirms WM-like (wlAS) properties of L6 astrocytes. **f**, The 3D model showing astrocyte area and layer heterogeneity. Astrocyte layering is regionally specialized across the dorsoventral and rostrocaudal extent of the cortex ($n = 1$ mouse, 10 tissue sections independently imaged).

to obtain regional/qualitative and cellular/quantitative information from in situ hybridization reintegrated into 2D tissue maps. In the present study, we focused on image acquisition from sections representing the mouse cerebral cortex and show that use of LaST map is feasible in human cortex for gene expression analysis at the single-cell level. Future elaboration with tissue warp integration would enable in silico three-dimensional (3D) map construction[35].

Analysis of cortical lamination mutants suggested that neuronal cues are required, directly or indirectly, to establish astrocyte laminae, particularly in L4 where neurons and mid-layer astrocyte layers do match most closely. Although the results in *Reeler* mice are consistent with astrocyte layering following that of excitatory neuronal layering, we cannot rule out that a cortical 'pre-pattern' instructs astrocyte precursors to respond to neuronal signals or other mediator cells involved in astrocyte specification. Indeed, astrocyte-encoded gradient patterns are typical of cells that have undergone polarization in response to a morphogen, and upstream regulation of this process is a subject for further study.

Together, our findings indicate the diversification of astrocytes in layer patterns that diverge from excitatory neurons, indicating a

composite 'neuroglial' cortical architecture with higher-order complexity. The present study has revealed markers for astrocyte cortical layers that could provide insight into specialized functions in development and/or disease[36].

Online content

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Methods

Mice. All mouse strains were maintained at the University of California San Francisco (UCSF), specific pathogen-free animal facility, and all animal protocols were approved by and in accordance with the guidelines established by the Institutional Animal Care and Use Committee and Laboratory Animal Resource Center. The day of birth was designated as P0.

Unless otherwise started, wild-type Swiss Webster mice were used for histology and spatial transcriptomics. *Aldh1l1-GFP* transgenic mice were generated by the GENSAT project[37]. *Emx1*^{cre} mice were obtained from the Jackson Laboratory (JAX, catalog no. 005628)[38]. *Satb2*^{flox} mice were a gift from Dr. Ralph Marcucio at UCSF[27]. Conditional knockouts were generated by breeding *Emx1*$^{cre/+}$;*Satb2*$^{flox/+}$ with *Satb2*$^{flox/flox}$ mice to obtain mutants (*cre/+;flox/flox*) and littermate controls (*cre/+;flox/+*). *Reeler* (*Reln/Reln*) mice were a gift from Dr. Eric Olson at SUNY Upstate University (JAX, B6C3Fe a/a-Reln). *Reln/*$^{+}$ mice were used as littermate controls. All mice were maintained on a mixed background.

Mouse tissue preparation. Mice were transcardially perfused at P14 or P56 with ice-cold phosphate-buffered saline (PBS) and 4% paraformaldehyde (PFA) in 1× PBS. Brains were dissected and post-fixed in 4% PFA for 24 h at 4 °C. Post-fixed brains were cryoprotected in 30% sucrose for 48 h at 4 °C and embedded in optimal cutting temperature compound (Tissue-Tek). Cryosections (16 µm) were collected on superfrost slides (VWR) using a cryostat (Leica, CM3050S) and stored at −80 °C.

Human tissue. Human brain tissue was obtained with informed consent under protocol 16/LO/2168 approved by the National Health Service's Health Research Authority at Addenbrooke's Hospital. Adult brain tissue biopsies were taken from the site of neurosurgery resection for the original clinical indication. For the purposes of the present study, samples were taken from pericontusional areas in traumatic brain injury (frontal cortex), lobectomy in epilepsy surgery (temporal cortex) and peritumoral tissue (temporal cortex).

Tissue specimens were collected in Addenbrooke's Hospital and transferred to a CL2 facility where it was processed. Tissue was dissected and fixed in 4% PFA for 48–72 h. Once fixed, samples were placed in 20% sucrose for cryoprotection for 24–48 h, mounted on optimal cutting temperature compound and stored at −80 °C.

Additional human brain tissue was collected in a de-identified manner with previous patient consent in strict observance of the legal and institutional ethical regulations of the UCSF Committee on Human Research. Protocols were approved by the Human Gamete, Embryo and Stem Cell Research Committee (institutional review board) at UCSF. For the present study, one postmortem sample was taken from the superofrontal gyrus and processed as above. Human brain blocks were cryosectioned to 16 µm.

SmFISH assay design and probes. Mouse and human tissue smFISH was performed using the RNAScope LS Multiplex Assay (Advanced Cell Diagnostics (ACD))[39]. In this assay, the smFISH SNR is amplified by branched-DNA complexes formed on target transcripts and tyramide signal amplification (TSA)-based labeling. Target RNAs are initially hybridized to a series of single-stranded DNA '*z*-probes'. Each *z*-probe is composed of (1) a 18- to 25-nucleotide region complementary to the target RNA, (2) a spacer sequence and (3) a 14-nucleotide tail region. These probes are tagged by branched DNA amplification trees: pairs of *z*-probes are hybridized to oligo-preamplifiers, across their bridged tail sequences, which are then tagged by 20 oligo-amplifiers (see Fig. 1a). Each oligo-amplifier is labeled with 20 horseradish peroxidase (HRP) enzyme molecules. In general, a 1-kilobase region on the target transcript is hybridized by 20 *z*-probe pairs in tandem, which can yield up to 8,000 HRP labels per target. The fluorescent smFISH signal is consequently generated by the addition of tyramide-conjugated fluorophores. Tyramide is enzymatically converted into a highly oxidized intermediate by HRP, which covalently binds to the proteins at or near the HRP label, depositing a large number of fluorophores for probe detection[40]. The combination of branched-DNA and TSA amplification notably boosts the sensitivity and SNR of the RNAScope assay, allowing fast confocal imaging of large tissue areas with short exposure times (see Image analysis).

Multiplexed detection. To achieve 4-plex transcript detection with RNAScope, target *z*-probes are assigned to one of four different channels (C1–C4) that contain distinct tail sequences. Tissue samples are hybridized to the mixture of C1–C4 probes, followed by generation of channel-specific amplification trees. Finally, probes are sequentially developed with TSA through incubation cycles of channel-specific HRP labels, tyramide-conjugated fluorescent dyes and chemical enzymatic quenchers.

Probe information. All of the RNAScope probes used in the present study and relevant information including target sequences are listed in Supplementary Table 3. Further information is readily available from the manufacturer (https://acdbio.com/catalog-probes). To assess the background signal from the assay, target probes against the bacterial *DapB* mRNA were used as negative controls. Target probes against mouse and human housekeeping genes were used as positive controls. In the mouse cortical astrocyte expression screen, the *Glast* probe was always assigned to C4 and multiplexed with other probes in the C1–C3 channels. With human tissue, the *Glast* probe was assigned to C3 and multiplexed with C1–C2.

Automated smFISH and IHC. All histology on mouse and human brain cryosections was automated on a Leica BOND RX robotic stainer after manual baking and dehydration. Tissue sections were first processed for three- or four-gene smFISH using the RNAScope LS Multiplex Assay (ACD). After smFISH, antibody staining was performed using TSA and slides were manually coverslipped for imaging.

Baking and dehydration. Tissue cryosections were removed from −80 °C and thawed at room temperature for 15 min. Samples were then baked at 65 °C for 45 min in a vertical position on a slide holder (Tissue-Tek) in an oven. After baking, samples were dehydrated in a series of 50%, 70%, 100% and 100% ethanol (5 min each) in staining dishes (Tissue-Tek) and air-dried for 10 min before automated RNAScope.

Automated histology design and setup. For use on the Leica BOND RX, all histology consumables were transferred to barcoded reagent containers. Staining protocols that list the order and durations of reagent incubations and washes were created on the BOND controller software. Slides were assigned unique barcode labels coupled to staining protocols and placed on to temperature-controlled pads on the instrument. Flow-through chambers were assembled across the whole slides using plastic coverplates. During staining, a liquid volume of 150 µl was dispensed to each slide on every step using automated liquid handling. Reagents were flushed at least once before the main incubations to ensure uniform coverage of the slide. Between reagent incubations, multiple short washes were performed. All incubations were performed at room temperature unless indicated otherwise. For the 4-plex RNAScope smFISH protocol, a maximum number of 20 slides could be processed against 10 different probe mixtures in a single run (for example, 40 different genes screened across 2 sets of biological replicates). The combined multiplexed RNAScope smFISH and IHC protocol for 20 slides ran overnight on the Leica BOND RX, lasting ~17 h.

Automated RNAScope smFISH. The RNAScope LS Multiplex Assay (ACD) was performed largely according to the instructions from the manufacturer and modifications are noted here. Full details of the protocols for 4-plex smFISH and the consumables used in the present study are provided in Supplementary Table 4.

Four-plex probe hybridization. To perform 4-plex RNAScope on mouse brain cryosections, samples were initially permeabilized with heat and protease treatment to improve probe penetration and hybridization. For heat treatment, P14 and P56 samples were incubated in BOND ER2 buffer, pH 9.0 (Leica) at 95 °C for 2 and 5 min, respectively. For protease treatment, P14 and P56 samples were incubated in ACD protease reagent at 42 °C for 10 and 15 min, respectively. Before probe hybridization, samples were incubated in hydrogen peroxide for 10 min to inactivate endogenous peroxidases and ACD protease. After pre-treatment, samples were incubated in target *z*-probe mixtures (C1–C4) for 2 h at 42 °C. The C2–C4 probes are provided at 50× concentration by ACD and were diluted 1:50 in C1 probes. As an exception, the C4 probe for the high-expressing *Glast* mRNA was used at 1:100 in the astrocyte screen for reagent conservation.

SmFISH signal amplification. After probe hybridization, branched DNA amplification trees were generated through sequential incubations in AMP1, AMP2 and AMP3 reagents for 15–30 min each at 42 °C with high-stringency washes with LS Rinse buffer between incubations. After amplification, probe channels were detected sequentially via HRP–TSA labeling. To develop the C1–C3 probe signals, samples were incubated in channel-specific HRP reagents for 15 min at 42 °C, TSA fluorophores for 30 min and HRP-blocking reagent for 15 min at 42 °C. The probes in C1, C2 and C3 channels were labeled using Opal 520, 570 and 650 fluorophores (Perkin Elmer, diluted 1:2,500) respectively. Finally, to develop the C4 probe, the Atto425 fluorophore was used for six-color imaging on the Operetta system. The C4 probe complexes were first incubated with TSA–biotin (Perkin Elmer, 1:500) for 30 min, followed by streptavidin-conjugated Atto425 (Sigma, 1:400) for 30 min. Multiple short washes were performed between incubations throughout the protocol using the BOND Wash buffer (Leica) and deionized water (full protocol listed on Supplementary Table 4).

Three-plex smFISH. To perform 3-plex RNAScope on human brain cryosections, samples were heat treated for 10 min and incubated in protease for 15 min. Probe hybridization and branched DNA amplification were performed as described above. To develop C1–C3 probes, Opal fluorophores (520, 570 and 650) were used at a lower dilution (1:300) due to higher autofluorescence on postnatal human brain sections. To distinguish RNA spots from lipofuscin autofluorescence, spots that appear identical across Opal 520 and 570 channels were filtered out.

Automated IHC. RNAScope smFISH was directly followed by antibody staining for the neuronal marker NeuN on the BOND RX system. To improve antibody staining after IHC and perform six-color imaging on the Operetta, the NeuN signal was amplified using TSA–biotin and the Alexa-700 fluorophore. Samples were first blocked in antibody-blocking solution (Perkin Elmer) for 20 min. To block any

available TSA–biotin sites from the smFISH assay, samples were incubated in 0.2% avidin (Sigma) for 20 min and 0.05% biotin (Sigma) for 30 min. After the avidin–biotin block, samples were incubated in chicken anti-NeuN antibody (Millipore) diluted 1:500 in blocking solution for 1 h. To develop the antibody signal, samples were incubated in goat anti-chicken HRP (Thermo Fisher Scientific, 1:500) for 1 h, TSA–biotin (1:200) for 10 min and streptavidin-conjugated Alexa-700 (Sigma, 1:200) for 30 min. After antibody staining, samples were incubated in DAPI (Sigma, 0.25 μg ml^{-1}) to mark cell nuclei, and washed multiple times in deionized water. After final washes, slides were briefly air-dried and manually mounted using ~170 μl of Prolong Diamond Antifade (Fisher Scientific) and standard coverslips (24 × 50 mm^2; Fisher Scientific). The full IHC protocol is listed in Supplementary Table 4.

Automated spinning-disk confocal imaging. Tissue sections were imaged on an Operetta CLS high-content screening microscope (Perkin Elmer). To perform six-color smFISH–IHC imaging, this system was equipped with eight light-emitting diode light sources, ×5 air and ×40 water objectives, wide-field and spinning-disk confocal imaging modules, and narrow-band emission filters. The fluorophores, light sources, exposure times and emission filters used for mouse and human tissue-imaging experiments are listed in Supplementary Table 5. Image acquisition and analysis were controlled using the Harmony software (Perkin Elmer).

Tissue identification. To locate whole-tissue sections or regions of interest for high-resolution imaging, entire slides were initially scanned under low magnification in wide-field mode. Each slide was imaged for nuclear DAPI and NeuN staining, if applicable, using a ×5 numerical aperture (NA) 0.16 objective (pixel size: 7.2 μm) under 5 min. To automatically locate the *xy* coordinates of tissue sections, a Harmony analysis script was used to detect DAPI$^+$ areas. Whole-slide DAPI images were stitched, smoothed with Gaussian blurring and analyzed with a global threshold. The detected DAPI$^+$ areas were size filtered to remove staining artifacts and slightly expanded to ensure complete tissue coverage. The resulting areas were used to automatically set the *xy* field positions of the subsequent ×40 scan. Alternatively, regions of interest for ×40 scans were manually selected on low-magnification previews. Selected ×40 fields were imaged with a 7% overlap.

Confocal imaging. The high-resolution smFISH images of tissue sections were acquired on the spinning-disk confocal mode using a sCMOS camera and a ×40 NA 1.1 automated water-dispensing objective. The field of view was 320 × 320 μm^2 and the pixel size was 298 nm. A P14 mouse brain hemisection comprised 200–300 fields depending on its anatomic position. Each field was imaged as a *z*-stack consisting of 20–30 planes, with a 1-μm step size across each color channel. An infrared laser was used to autofocus on the position of the coverslip and the relative *z*-heights of tissue sections were manually identified by imaging DAPI on sample fields before tissue-wide scans. Each *z*-plane was imaged across 4–6 channels depending on the experiment, with exposure times for mouse smFISH channels between 60 and 120 ms (see Supplementary Table 5). The ×40 multi-channel settings and tissue heights were entered into an experimental layout on Harmony and automatically executed after low-magnification tissue identification scans.

Image analysis. To segment single neurons and astrocytes and quantify RNA spots from high-resolution images, analysis scripts were created on Harmony software (Perkin Elmer) using customizable, predefined, function blocks (italicized below). The complete single neuron and astrocyte segmentation pipelines are provided in Supplementary Tables 6 and 7. Each ×40 field was analyzed separately to optimize processing time of large datasets.

Quantification of neuronal gene expression in situ. Segmentation of neurons. Maximum intensity *z*-projection images were calculated across each channel to generate 2D images from *z*-stacks. NeuN$^+$ neurons were segmented in three steps: (1) supervised texture segmentation was performed at a coarse scale to locate NeuN$^+$ areas on images and filter staining artifacts on tissue sections (*find texture regions*). Intensity and size thresholding then identified the NeuN$^+$ neuronal soma (*find image region*). (2) Neuronal nuclei were segmented within the NeuN$^+$ soma from Gaussian blurred DAPI images using intensity, size and contrast thresholds (*find nuclei*). (3) The neuronal cytoplasm was segmented around the nuclei within the boundaries of the neuronal soma using NeuN intensity thresholding (*find cytoplasm*).

Filtering single neurons. Neuron segmentation yielded single neurons as well as doublets/triplets that overlap in *z*-projection images and neurons that are partially contained in tissue sections (see Supplementary Fig. 1b). To automatically distinguish single neurons, morphologic (for example, area, roundness) and intensity (for example, pixel sum over DAPI and NeuN) properties of segmented cells were measured and used to train a supervised linear classifier (*select population*). For the training set, we manually selected >100 single, doublet and partial cells across multiple tissue sections and cortical areas. The resulting classification was validated across the cortex by manual inspection of several fields.

RNA spot calling in neurons. RNA spots were identified by the detection of local intensity maxima across each smFISH channel in the neuronal soma (*find spots*). Individual spots were identified with an upper radius threshold of 750 nm.

The number of RNA spots per single neuron was calculated for each smFISH channel (*calculate properties*). Last, all DAPI$^+$ nuclei were identified across the given field for use in brain region segmentation (see Anatomical annotation of mouse cortical layers and areas).

Quantification of astrocyte gene expression in situ. Segmentation of astrocytes. Maximum intensity *z*-projection images were generated as above. Background illumination profiles of fluorescent channels were mapped to correct uneven illumination (*flatfield correction*). *Glast*$^+$ astrocytes were segmented in three steps: (1) supervised texture segmentation was performed at a fine scale to identify *Glast*$^+$ astrocyte cytoplasm and main processes (*find texture regions*). To train texture analysis, over 100 points were selected inside versus outside *Glast*$^+$ cortical astrocytes across several tissue sections. Astrocyte cell areas were then filtered by size to remove partial cells, and holes across astrocyte nuclei, which are weakly labeled by *Glast* smFISH, were filled (*select region*). (2) Astrocyte nuclei were segmented within *Glast*$^+$ cell areas from Gaussian blurred DAPI images using intensity, size and contrast thresholds (*find nuclei*). To remove false-positive, non-astrocyte nuclei that overlap with astrocyte processes in *z*-projections, additional morphology and *Glast* intensity filters were used (*select population*). (3) The astrocyte cytoplasm and processes were segmented around the nuclei within the cell areas using *Glast* intensity thresholding (*find cytoplasm*).

Filtering single astrocytes. To remove overlapping astrocyte doublets and partial cells, cells were filtered based on morphologic (for example, area, roundness) and intensity (for example, pixel sum over DAPI and *Glast*) properties. As shown in Supplementary Fig. 11, upper-layer astrocytes were slightly larger, consistent with previous reports[7], and showed higher *Glast* levels than those in deep layers. This analysis identified similar numbers of astrocytes across cortical areas in technical replicates (see Supplementary Fig. 11). Manual validation of this pipeline across ten tissue sections from two biological replicates covering multiple cortical areas (somatosensory, visual, auditory) showed that >90% of astrocytes were correctly identified using this pipeline.

Given their intimate cell–cell interactions, astrocyte processes occasionally overlapped with neuronal soma and other nuclei in maximum *z*-projection images. To remove overlapping non-astrocyte nuclei, the DAPI signal in the astrocyte cytoplasm was identified with intensity thresholding and subtracted from the *Glast*$^+$ cell area (cytoDAPI-filtered cells, see Fig. 2c). To remove overlapping neurons, the NeuN signal was used to segment neurons as above. Astrocytes that notably overlap with neurons were discarded (>50% overlap between the astrocyte nuclei and neuronal soma). NeuN$^+$ neurons were then subtracted from the cytoDAPI-filtered *Glast*$^+$ cell area (see Fig. 2c), resulting in astrocytes filtered against overlapping neurons and nuclei.

RNA spot calling. RNA spots were quantified across single astrocytes as elaborated for neurons previously. In addition to the filters used above, cells that are high expression outliers (>99.5% of spot counts per gene across the brain) were also filtered out. RNA spots in neurons, and cytoDAPI-filtered and double-filtered astrocytes were also quantified.

Data processing. The analysis was performed on a desktop workstation with two 6-core Intel i7–4930K 3.4 GHz CPUs and 64 GB of RAM. The neuronal dataset shown in Fig. 1, consisting of 10 tissue sections and ~300,000 images, was analyzed for 6 h. The data were exported from Harmony as a single-cell matrix showing cell coordinates, morphologic and intensity measurements, and RNA spot counts per cell. Brain region segmentation was performed in MATLAB (described in "Manual segmentation of brain regions" below). Data organization and plotting were done in R.

Anatomic annotation of mouse cortical layers and areas. For mapping neuron and astrocyte subtypes across the cortex, one P14 mouse brain hemisphere was sectioned along the coronal plane to generate an eight-slide series containing ten sections each. One slide was used to map the expression of cortical layer neuron markers (see Fig. 1) and the remaining slides were assayed with layer astrocyte markers (see Figs. 3 and 6). The cortical layers and areas were annotated using NeuN and DAPI staining as well as layer neuron marker expression patterns as anatomic landmarks (see Supplementary Fig. 7 for areas). The Paxinos (P6, plates 9–40) and Allen Mouse Brain ISH Atlases (P56, sections 38–88) were used as anatomic references. Cortical areas were annotated broadly across the anteroposterior and dorsoventral axes, grouping functionally related areas (for example, the anterior division of the somatosensory cortex contains the mouth and limb areas). The list of cortical area abbreviations and groupings is given in Supplementary Table 8.

For screening layer astrocyte gene expression (see Fig. 2), 2 P14 mouse brain hemispheres (biological replicates) were sagittally sectioned to generate 18 slides. Each slide contained four sections through the somatosensory cortex from each replicate, corresponding to the areas used for RNA-seq profiling of layer astrocytes.

For examining cortical layers in *Satb2* cKO and *Reeler* mice (see Fig. 5), three coronal sections through the somatosensory barrel cortex were collected from littermate control and mutant brains ($n = 3$ biological replicates each). Each slide contained sections from one control and one mutant brain.

To normalize the layer depth of neurons across cortical areas, we automatically measured the normalized distance between individual neurons, cortical pia and WM using the SlideSegmenter application below.

Manual segmentation of brain regions. Brain areas were manually segmented on low-magnification (×5) images of DAPI/NeuN-stained brain sections. These segmentation masks were overlaid on *xy* coordinates of high-magnification (×40) scans to annotate single cells. The offset between ×5 and ×40 objectives was manually corrected by aligning DAPI$^+$ nuclei (identified across all cells in each field at the end of the segmentation pipelines). To stitch images from Harmony, draw and name segmentation masks, align low–high magnification data and perform batch segmentations, the SlideSegmenter application was created to work on the MATLAB environment and made publicly available (https://bitbucket.org/alexmatlab/slidesegmenter).

Identification of neuronal subtypes from smFISH data. To identify cortical neuron subtypes in an unbiased manner from smFISH data, we adopted the following workflow.

Filtering and normalization. For downstream analysis, neurons were (1) selected from the eight broad cortical areas that show the full complement of layers with respect to the four genes profiled and (2) filtered with a minimum cumulative spot-count threshold of 20.

Spot counts then had a value of 1 added and were log transformed (log (spot counts + 1)).

Clustering. Clustering was performed for cells from each region individually with the four genes profiled using graph-based clustering implemented by Seurat (FindClusters function, resolution 0.5)[41]. Briefly, a *K*-nearest neighbor graph based on Euclidean distance is constructed from the expression values for each cell. Edges between cells were weighted, based on shared overlap in neighborhoods determined by the Jaccard distance. Cells were iteratively grouped together with the goal of optimizing the density of links inside communities compared with links between communities.

t-SNE. For visualization, *t*-distributed stochastic neighbor embedding (*t*-SNE) coordinates were calculated from the expression values for each cell (independent of the clustering) using perplexity 250 with Seurat (RunTSNE function). The *t*-SNE plots were then colored by the cluster assignments derived above, gene expression values or other features of interest.

Hierarchical clustering. The mean expression profiles of each of the Seurat clusters derived from each brain region were taken and hierarchically clustered together, based on Euclidean distance using Ward.D2 clustering (hclust(dist, method = "ward.D2") R function). The resulting dendrogram was then cut at height 1.9 yielding 18 groups (cutree(hc, h = 1.9) R function). These groups were manually annotated to 10 major subtypes based on high expression differences (see Supplementary Fig. 5) and similarity among the spatial distributions of identified groups.

Cortical layer and purified layer astrocyte RNA-seq. *Cortical layer dissection.* *Aldh1L1-GFP*$^+$ mice were transcardially perfused at P14 with ice-cold HBSS to wash away the blood. Brains were dissected and cortical hemispheres were cut sagittally on a vibratome in ice-cold HBSS to 300-µm-thick sections. Sections from eight littermate pups were pooled for each experiment ($n = 3$ biological replicates for astrocyte purification and $n = 2$ biological replicates for whole-layer RNA extractions) and microdissected to separate upper (L2–4) and deep (L5–6) cortical layers. The L4 of the somatosensory barrel cortex, which appears as dark barrels separated by light septa under bright-field illumination, was used as an anatomic landmark for layer microdissections (see Supplementary Fig. 8). To prevent contamination with WM astrocytes, the most superficial layers that contain pial and L1 astrocytes, and the deep subcortical WM that contains fibrous astrocytes, were discarded. For each experiment, the dissections were completed in under 90 min and the tissue was kept in ice-cold HBSS.

Flow cytometry. To purify cortical layer astrocytes, tissue dissociation was performed as described previously[17]. Briefly, cortical layer tissue was minced with a forceps and enzymatically dissociated with papain (20 U ml^{-1}) in dissociation buffer (glucose 22.5 mM, ethylenediaminetetraacetic acid 0.5 mM, phenol red), L-cysteine (1 mM) and DNase (125 U ml^{-1}) for 80 min at 33 °C. Tissue was then washed in inhibitor solution (dissociation buffer with ovomucoid (1.0 mg ml^{-1})) and centrifuged for 5 min at 200*g*. Supernatant was discarded, the tissue was resuspended in the inhibitor buffer and mechanically disrupted using a P1000 pipette. Dissociated cells were layered on to inhibitor buffer with concentrated ovomucoid (5 mg ml^{-1}) and centrifuged for 5 min at 200*g*. Finally, the cell pellet was resuspended in staining medium with DAPI. *Aldh1l1-GFP*$^+$ and *Aldh1l1-GFP*$^-$ cells were sorted as previously described[8] on a BD FACS Aria II and gated on forward/side scatter, live/dead by DAPI exclusion, and GFP, using GFP and DAPI controls to set gates for each experiment (see Supplementary Fig 8).

RNA-seq and analysis. Total RNA from FACS-purified cortical layer astrocytes and whole cortical layers was extracted with Trizol LS (Invitrogen) and purified using the RNeasy Kit (Qiagen). Complementary DNA was generated from full-length RNA using the NuGEN RNA-Seq v.2 kit that employs the single primer isothermal amplification method to deplete ribosomal RNA, and sheared by Covaris to yield uniformly sized fragments. RNA-seq libraries were generated using the NuGen Ultralow kit for adapters, barcoding and amplification, and purified using the Agencourt XP magnetic beads, quality controlled with an Agilent bioanalyzer and quantified by quantitative PCR.

Five libraries were pooled per lane across three lanes for single-end (SE50) sequencing on a HiSeq 4000. Read quality was assessed using FastQC (v.0.11.4) and five nucleotides at the 5′-end were trimmed; 45-nucleotide long reads were aligned to the mouse reference genome (Ensembl GRCm38) using TopHat2 (v.2.0.11)[42]. The multiple hit parameter (−*g*) was set to 1 to exclude reads with multiple genomic alignments. On average, 68 million reads were uniquely mapped to each sample (range 59–87 million reads). Read counts per gene were calculated using SAMtoools (v.0.1.19)[43] and HTSeq (v.0.6.1p1) with default parameters[44]. DESeq2 (ref. [45]) was used to detect differentially expressed genes among upper- and deep-layer astrocytes ($n = 3$ biological replicates) and whole cortical layers ($n = 2$ replicates). Purified deep-layer astrocytes showed low levels of contaminating oligodendrocyte marker gene expression (for example, *MBP*); these genes were excluded from analysis using a mild astrocyte-specific expression threshold (astrocyte versus whole-layer expression >0.1). To identify the top differentially expressed genes between upper and deep gray matter astrocytes, an expression threshold of 5 FPKM (fragments per kilobase per million reads) was used with an FDR < 0.05. The resulting list of 159 differentially expressed, layer astrocyte genes is provided in Supplementary Table 9.

Spatial reconstruction of astrocyte layer heterogeneity. *Pre-processing of astrocyte scRNA-seq data.* ScRNA-seq data of P56 cortical astrocytes were obtained from Batiuk et al.[22], who used a modified version of the Smart-seq2 protocol. The number of unique molecules was estimated from raw read counts using the Census method from the Monocle package[46]. The unique molecule version of the data was normalized with size factors according to standard scatter workflow[47].

Spatial reconstruction. We reconstructed spatial profiles of genes measured with scRNA-seq using a published method[23]. Implementation of the method in MATLAB was provided by S. Itzkovitz (see the original publication for the description of implementation details). Here we summarize the method.

Constructing the prior on the number of reads for marker genes in scRNA-seq cells. Molecule counts of genes from the processing step of image analysis were fit by a gamma distribution for each gene *i* and spatial bin *j*:

$$\mathrm{sc}_{ij} = \mathrm{gamma}(\mathrm{os}_{ij}, \mathrm{or}_{ij}) \quad (1)$$

where sc is spot counts, os is the observed shape of gamma distribution and or is the observed rate.

To match this distribution to the statistical properties of scRNA-seq data and produce the prior distribution of scRNA-seq reads for each gene *i* and spatial bin *j*, the following correction was applied for each scRNA-seq cell *c*:

$$f_c = \left(\frac{s_c}{\mathrm{sm}}\right) \quad (2)$$

where *f* is the distribution rescaling factor, *s* is the scRNA-seq sampling constant and sm is the smFISH sampling constant;

$$p_{ijc} = \frac{f_c}{(\mathrm{or}_{ij} + f_c)} \quad (3)$$

where *p* is the negative binomial probability;

$$\mathrm{prc}_{ijc} = \mathrm{negativebinomial}(r = \mathrm{os}_{ij}, p_{ijc}) \quad (4)$$

where prc is the prior on expected scRNA-seq reads for each marker gene, and the bin and cell smFISH sampling factor reflects the expected proportion of the total number of transcripts in a cell captured in smFISH images (we used 0.15). The scRNA-seq sampling factor reflects the proportion of total number of transcripts (approximated as 180,000) captured by scRNA-seq for each cell measured. The total number of transcripts was chosen based on Halpern et al.[23] who examined haptocytes: 180,000 is roughly 1 in 4 of the hepatocyte molecule number—1 in 2 to account for tetraploidy and 1 in 2 to account for smaller cell size of astrocytes. In practice, sampling levels are computed for 8 bins of cells with similar sampling levels (for example, 0–0.1%, 0.1–0.3%) to speed up the Monte Carlo sampling used to construct the prior. We used five spatial bins (*j*) for equally sized intervals of cortical depth.

Using the prior to assign cells to spatial bins. To find the probability of each scRNA-seq cell coming from spatial bin *j* given the expression of spatial markers, we used Bayes' formula:

$$P(\mathrm{Bin}_j|\mathrm{gene}_i \text{ reads}) = \frac{P(\mathrm{gene}_i \text{ reads}|\mathrm{prc}_{ijc}) \times P(\text{cells in Bin}_j)}{\sum_{\text{bin } j=1}^{\text{bin } 5} P(\mathrm{gene}_i \text{ reads}|\mathrm{prc}_{ijc}) \times P(\text{cells in Bin}_j)} \quad (5)$$

where $P(\text{cells in Bin}_j)$ is the proportion of cells in spatial bin j, and the prc prior on reads comes from Equation (4).

The method assumes that expression of different genes is independent and the probability of assignment to spatial bins is computed as:

$$P(\text{Bin}_j|\text{gene}_{i-n}\text{ reads}) = \prod_{\text{gene } i=1}^{\text{gene } n} P(\text{Bin}_j|\text{gene}_i\text{ reads}) \quad (6)$$

This results in a posterior probability matrix of cells by bins. A probability weight matrix is obtained by normalizing the column sums to one.

Reconstructing expression of genes across spatial bins. The probability weight matrix, P, and the scRNA-seq expression matrix, E, are combined to obtain spatial profiles (S) of a genes-by-bins matrix as a weighted average (see Supplementary Tables 10 and 11):

$$S = E \times P \quad (7)$$

Identifying notably zonated profiles. To find which profiles are notably zonated, we assigned cells to the spatial bin with the highest posterior probability. The Kruskal–Wallis test was used to find which genes are significantly different between cells assigned to different spatial bins.

Leave-one-out validation. To benchmark the model, we used leave-one-out analysis (see Supplementary Fig. 18). We used 15 genes with smFISH profiles to reconstruct one left-out gene and compared the reconstructed profile with the average expression observed with smFISH. To measure the similarity of predicted and observed profiles, we used Kullback–Leibler divergence, based on the relative differences in expression between spatial bins rather than the absolute number of genes per bin.

Manual IHC. Cryosections from *Aldh1L1-GFP* mice were manually stained for GFP, astrocyte and neuron marker antibodies. Samples were subjected to heat-induced antigen retrieval in 10 mM sodium citrate, pH 6 buffer for 5 min at 75 °C, then permeabilized and blocked in 10% goat serum in 1× PBS with 0.1% Triton X-100 (PBST) for 1 h. Primary antibodies were diluted in the blocking solution and incubated overnight at 4 °C. After multiple PBST washes, samples were incubated in secondary antibodies and DAPI diluted in blocking solution for 1 h. After PBS and deuterated water washes, samples were mounted using Fluoromount-G (SouthernBiotech). The following primary antibodies were used: chicken GFP (Aves, GFP-1020, 1:2,000), mouse NeuN (Millipore, MAB377, 1:500), rabbit GS (Sigma, G2781, 1:2,000) and rabbit pSmad (Cell Signaling, 13820S, 1:200). Goat secondary antibodies conjugated to Alexa fluorophores (Molecular Probes) were used for labeling. The *Aldh1L1-GFP* samples were imaged on a Leica TCS SPE laser confocal microscope with a ×40 oil immersion objective.

Statistics. The violin plots in Fig. 1d, Supplementary Figs. 2b,c, 11c and 21c, and the box plots in Supplementary Fig. 20a,b were plotted to show the data distribution as follows: the lower, middle and upper hinges correspond to the 25th, 50th (median) and 75th percentiles. The whiskers show data points within 1.5 times the interquartile range (distance between the first and the third quartiles) of the 25th and 50th percentiles.

No statistical methods were used to predetermine sample sizes, but our sample sizes were similar to those reported previously[10–12]. The smFISH data (RNA spot counts per cell) and scRNA-seq data distribution were assumed to be gamma-Poisson but this was not formally tested.

For layer-switch experiments (see Fig. 5), data collection and analysis were performed blind to the genotypes of the animals. Animals were randomly allocated to experimental groups. Two-tailed Student's *t*-tests were used to assess statistical significance in layer-switch experiments (see Fig. 5).

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The raw bulk RNA-seq data are available at the Gene Expression Omnibus under the accession code GSE140822. The scRNA-seq data will be made available under https://holt-sc.glialab.org/sc. Other data are available as Supplementary Materials or from the corresponding author upon request.

Code availability

The code for spatial reconstruction of single-cell astrocyte RNA-seq can be found at https://github.com/vitkl/cortical_astrocyte_mapping. The SlideSegmenter code is available at https://bitbucket.org/alexmatlab/slidesegmenter/src/master. The Harmony image analysis scripts are provided as Supplementary Materials. Other code is available upon request.

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Author contributions

O.A.B. and D.H.R. conceived the study. O.A.B. planned the experiments. O.A.B., T.B., S.H. and K.R. performed histology and imaging experiments. O.A.B. performed the image analysis. K.P. and J.S. contributed to the histology and imaging pipeline. D.P. and O.A.B. analyzed neuron gene expression data to identify subtypes. A.B. wrote the SlideSegmenter software. A.M.H.Y., P.H., M.F.P. and E.H. provided human tissue. O.A.B. and L.B.H. performed layer astrocyte purification. O.A.B. and R.K. analyzed RNA-seq data. G.S., D.H.G. and E.M.U. supervised analysis of neuron smFISH and astrocyte RNA-seq data. A.M., M.B. and M.G.H. generated the astrocyte scRNA-seq data. V.K. and M.H. analyzed the scRNA-seq data and performed spatial reconstruction analysis. K.S. and S.C. supported mouse work and genotyping. O.A.B. and D.H.R. wrote the manuscript with feedback from all authors.

Competing interests

The authors declare no competing interests.

Additional information

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	BOND RX 5.2 controller software was used for controlling the Leica BOND automated stainer. Harmony versions 4.5 and 4.9 (Perkin Elmer) for image acquisition. Harmony image acquisition scripts are available upon request. Leica Application Suite X (LAS X) was also used for image acquisition.
Data analysis	Image data analysis was performed on Harmony, MATLAB v2017 and R v3.5.1. The harmony image analysis scripts were exported to supplementary tables 6 and 7. MATLAB code for slideSegmenter is provided at https://bitbucket.org/alexmatlab/slidesegmenter/src/master/. MATLAB and R code for spatial reconstruction of gene expression is provided at https://github.com/vitkl/cortical_astrocyte_mapping. All other R code is available upon request. Tophat2 (v2.0.11), SAMtools (v0.1.19), HTSeq (v0.6.1p1), DESeq2 and Seurat (v2.2.1) were used to analyze RNAseq data.

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- A list of figures that have associated raw data
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The raw bulk RNA-sequencing data (figure 2) are available at the Gene Expression Omnibus (GEO) under the accession code GSE140822. The single cell RNA-

sequencing data (Figure 4) will be made available under https://holt-sc.glialab.org/sc/ that will include a GEO link. Other data are available as Supplementary Materials or from the corresponding author upon request.

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Life sciences study design

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Sample size	Tissue sections from two to three mice were used in the experiments in Figures 2 and 5. Multiple tissue sections were analyzed from single mouse brains for large mapping experiments reported in Figures 1,3,4 and 6. The precise number of animals and tissue sections is reported in the figure legends and methods. Three human samples were used. No statistical methods were used to pre-determine sample sizes but our sample sizes are similar to those reported previously (references 10-12).
Data exclusions	Single neuron smFISH data was filtered to remove cells with cumulative spot counts under 20. Single astrocyte smFISH data was filtered to remove expression outliers with cumulative spot counts over 99.5% of all astrocytes.
Replication	Histology and imaging experiments were replicated on tissue sections from different animals (biological replicates) or multiple tissue sections across different anatomical levels of cortical areas. Independent experiments were performed on different days. Neuronal and astrocyte layer/area expression patterns were quantified in single wild-type animals in figures 1,3,4 and 5 and also observed qualitatively in a second set of biological replicates. In addition, these patterns were reproduced in 6 wild-type animals of two different genetic backgrounds used as controls in Figure 5.
Randomization	Animals were randomly allocated to experimental groups.
Blinding	For layer switch experiments (Figure 5), data collection and analysis were performed blind to the genotypes of the animals. Other experiments sued wild type animals.

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We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

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☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology
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Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	The following primary antibodies were used: chicken GFP (GFP-1020, Aves, 1:2000), mouse NEUN (MAB377, Millipore, 1:500), rabbit GS (G2781, Sigma, 1:2000) and rabbit pSmad (13820S, Cell Signaling, 1:200). Goat secondary antibodies conjugated to Alexa fluorophores (Molecular Probes) were used for labeling.
Validation	All antibodies were validated by vendors on mouse tissues for IHC with the information available on vendor websites.

Animals and other organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research

Laboratory animals	Unless otherwise started, wild-type Swiss Webster mice were used for histology and spatial transcriptomics. Aldh1l1-GFP transgenic mice were generated by the GENSAT project (37). Emx1-cre mice were obtained from The Jackson Laboratory (JAX #005628) (38). Satb2-flox mice were a gift from Dr. Ralph Marcucio at UCSF (27). Conditional knockouts were generated by breeding Emx1-cre/+; Satb2-flox/+ with Satb2-flox/flox mice to obtain mutants (cre/+; flox/flox) and littermate controls (cre/

	+;flox/+). Reeler (Reln/Reln) mice were a gift from Dr. Eric Olson at SUNY Upstate University (B6C3Fe a/a-Reln; JAX). Reln/+ mice were used as littermate controls. All mice were maintained on a mixed background.
Wild animals	No wild animals were used.
Field-collected samples	The study did not involve samples collected from the field.
Ethics oversight	All mouse strains were maintained at the University of California, San Francisco (UCSF) specific pathogen-free animal facility and all animal protocols were approved by and in accordance with the guidelines established by the Institutional Animal Care and Use Committee and Laboratory Animal Resource Center.

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Methodology

Sample preparation	Mouse cortical layer tissue were minced with a forceps and enzymatically dissociated with papain (20 U/mL) in dissociation buffer (glucose 22.5 mM, EDTA 0.5 mM, phenol red), L-cysteine (1 mM) and DNase (125 U/mL) for 80 min at 33oC. Tissue was then washed in inhibitor solution (dissociation buffer with ovomucoid (1.0 mg/mL)) and centrifuged for 5 min at 200 g. Supernatant was discarded, the tissue was resuspended in the inhibitor buffer and mechanically disrupted using a P1000 pipette. Dissociated cells were layered onto inhibitor buffer with concentrated ovomucoid (5 mg/mL) and centrifuged 5 min at 200 g. Finally, the cell pellet was resuspended in staining medium with DAPI.
Instrument	BD FACS Aria II.
Software	FACS DIVA software (BD Biosciences) and FlowJo software (TreeStar).
Cell population abundance	Cell population abundance >98% purity of AldhIII-GFP+ astrocytes were achieved, checked using double sorting.
Gating strategy	Cells were gated on forward/side scatter, live/dead by DAPI exclusion, and GFP, using GFP and DAPI controls to set gates for each experiment. See Supplementary Figure 8C.

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.